Supplementary Table. Thematic Contributions of Abstracted Data

Preferred Communication Themes	Communication Behavior Subthemes	Data Contributing to Subthemes	Direct Quotations Describing Communication Behaviors
(1) Disclosure-facilitating communication	Using open ended questions about pain	Afrell[43]: Using a variety of open-ended questions led to success in the encounter, as patients were more likely to **open up and disclose**. This was mutually beneficial in that patients felt relief and comfort opening up to their PTs, and PTs were **better able to diagnose and treat** with information gained in these disclosures.	"The key questions offered a rich and rewarding dialogue which influenced not only the patients but also the physiotherapists...The questions gave comprehensive answers." (p. 284)
			"Open ended questions related to pain in focus groups; 1. What do you think your pains are caused by? 2. How did your pains change your life? Tell us about it. Tell us about parts of your life where the pain is not important and where it takes over. Describe situations. 3. Can you rely on your body? Tell us about it. Do you feel that your body is with you or against you? Tell us about it. Has it always been like that? 4. If pain could talk, what would it relate? 5. After the event, and now that you know what you feel like today, would you have acted differently if you had been given the chance to relive your life?" (p. 283)
			"'Our' physiotherapists saw that the key questions started a process. The patients opened up inner 'closed doors' and feelings arose even though this was sometimes painful." (p. 286)
			"The questions make the patient prepared to reflect together with the physiotherapist on his/her aching body and what it is like to live with it. The key questions have thus proved to be effective and to make dialogues about the body possible." (p. 287)
		Synnott[51]: Specific biopsychosocial training (Cognitive Functional Therapy, CFT) training shifted physiotherapists' communication style to being more open and unrestricted. This trend/shift was seen with training and with experience.	"After completing the CFT training, participants reported a shift in their communication style from a rigid structured approach to an open and unrestrictive style. Participants identified how an open communication style promoted an easiness and fluidity in the exploration of the cognitive, psychological and social dimensions with patients." (p. 218)
	Using open-ended formulations such as 'tell me about your troubles' or 'what's your story?' at the beginning of an encounter	Opsommer[48]: Successful communication encounters involved opening the visit with an open-ended question.	"physiotherapist B used an open question in the beginning of the encounter 'tell me about your troubles?'…" (p. 218)
		Synnott[51]: Physical therapists with more experience begin their consultations with more open-ended formulations to patients.	"Now I think I'm much more open-ended, so I kind of ask open ended questions like 'What's your story?' you know, or 'What brings you here?' (P1)" (p. 218)
	Using gaze and nodding as continuers	Opsommer[48]: Coupling gaze with nodding during a patient's turn at talk **facilitated disclosure** and continuation of their narrative.	"One way for gathering information from patients was by means of a very open initial question, minimal acknowledgements of the patient's talk and the use of gaze and nods as continuers." (p. 210)

Preferred Communication Themes	Communication Behavior Subthemes	Data Contributing to Subthemes	Direct Quotations Describing Communication Behaviors
	Asking about lifestyle	Cowell[45]: Part of a comprehensive biopsychosocial approach to diagnosis and treatment is learning about a patient's lifestyle and inviting them to disclose **how their lifestyle impacts their pain and activity**.	"I think my common route is trying to delve what their underlying fears are, if there's any other drivers, maybe more obvious drivers, so is there, is work an issue? So talking about work a little bit more. Anything going on at home? ... the stresses of lifestyle." (p. 115)
	Facilitating open lines of communication	Serpanou[50]: Open lines of communication and availability were considered important elements for **facilitating disclosure** among patients.	"My communication with the patients is open. Any time the patients can call me and tell me their problems." (p. 37 Table 3)
(2) Rapport-building communication	Developing rapport and trust	Askew[33]: Developing rapport was synonymous with **developing a relationship with the patient**; this was seen as an essential component of care and key advice for new physical therapists.	"I think that is the most important thing, the level of communication between the therapist and the patient from the start ... finding out what is happening with the whole patient. There is more to pain than just the physical, there is a lot of mental and emotional aspects that are tied into pain" (p. 51)
			"Over 80% of respondents replied that they would tell a new therapist to **develop a good initial relationship with the patient**" (p. 51)
		Pincus[49]: Communicating in a way that **establishes trust** was important to physical therapists.	"Clinicians reported investing time and effort to **create trust** and an atmosphere in which patients felt able to discuss their fears" (p. 70)
	Friendly, respectful communication	Afrell[43]: A large component of respectful communication involved listening.	"You have to take everybody for who they are, listen to them, and try to get into their world and work with them because if you don't then you will never be able to pull them out because you aren't listening." (p. 50)
		Del Baño-Aledo[46]: Friendly and respectful communication was both commonly experienced and preferred by patients.	"friendly and respectful behavior of physiotherapists was a predominant patient experience reported in all focus groups" (p. 76)
	Displaying positive emotion and humor	Gard[41]: Humor was considered an effective therapeutic instrument, preferred by patients.	"Positive emotions such as interest and joy were expressed by most informants in the interview situation, when treatments had been successful, as joyful contacts with colleagues, or in situations where humour was used as a therapeutic instrument." (p. 237)
	Individualizing communication	Cooper[44]: Tailored communication resulted in more **patient-centered care**.	"specific approaches were more useful for some than others, suggesting that tailoring communication to the individual's needs is important" (p. 246)
			"Written communication was also discussed, often in a negative manner, suggesting that care should be taken to issue information acceptable to the individual" (p. 246)
			"Listening, understanding and getting to know the patient, and allowing the patient to explain their problem and to question the physiotherapist"" (p. 247)

Preferred Communication Themes	Communication Behavior Subthemes	Data Contributing to Subthemes	Direct Quotations Describing Communication Behaviors
		Serpanou[50]: Individualizing and tailoring communication to the patient was important for **addressing individual needs in the physical therapy plan of care.**	"Physical therapists need to be prepared to address those obstacles and open communication channels with the patients and their families taking into account the individual's needs, capacities, and resources." (p. 39)
(3) Empathic communication	Using therapeutic, supportive touch	Askew[33]: Touch was considered a marker of and a way to communicate a strong and supportive relationship.	"The amount of time I spend with a patient and the amount of touch that I provide, and the amount of support I give them in saying, 'I'm gonna help you' **makes them believe they can get better**." (p. 52)
	Showing empathy	Askew[33]: Physical therapists' personal experiences with pain helped them communicate about pain more empathically.	"… I have had [personal] experience with some back pain, just occasionally nothing major. But it really puts into perspective how scary it can be when something is minor and is really painful … it really makes you appreciate what some other people are going through … it makes me understand better what they're feeling … I'm probably a little more sympathetic" … "can put some of my own personal experience in it, always being active in my years and so forth. I think I was probably one of these guys who blew off pain, kind of worked through it. So when I am working with my patients, I have to be careful not to take their pain lightly." (p. 45)
	Providing emotional support	Del Baño-Aledo[46]: Listening skills were key to showing emotional support to patients.	"Supportive care was perceived during dialog with therapists and by their willingness to listen to patients' viewpoints or concerns." (p. 76)
		Askew[33]: Physical therapists perceived emotional support to consist of listening, encouraging, working with, and not judging their patients.	"I guess what I have learned is that you can't sit back and judge people when you don't know what their story is. You have to take everybody for who they are, listen to them, and try to get into their world and work with them because if you don't then you will never be able to pull them out because you aren't listening. That's what these people need the most, they are looking for support, they need encouragement, they need somebody to work with them to help them." (p. 50)
		Cowell[45]: Reassurance and emotional support was valued and needed within the direct context of patients' physical therapy treatment but with their health as a whole as well.	" 'They need that reassurance that actually, there isn't anything going on, particularly if you get people that have had a family history of cancer or they know someone that's had surgery and ended up in a wheelchair… until they've been given that all clear you can't actually move them forward no matter how hard you try' " (p. 115)
			"physiotherapists felt that they required more time in the initial encounter to facilitate and support emotional disclosure" (p. 116)
	Providing early follow-up support	Cowell[45]: One unique benefit to physical therapy is the opportunity to provide early follow up; emotional support during these follow-ups was immensely beneficial to patients.	"early follow-up support was considered optimal for reinforcement and to reassure anxious patients, but was a luxury rarely afforded in primary care" (p. 116)
			"While 'our' physiotherapists appreciated that the key questions might make the patients open up, they sometimes also became frustrated. As

Preferred Communication Themes	Communication Behavior Subthemes	Data Contributing to Subthemes	Direct Quotations Describing Communication Behaviors
	Facing uncertainty when responding to patients' emotional displays	Afrell[43]: Physical therapists can experience difficulty responding to sad emotions elicited by patients through certain lines of questioning.	professionals they could experience a sense of powerlessness when confronted with too much sorrow and suffering." (p. 286)
			"This distancing may offer therapists a way to protect themselves, but it can make the patient feel dismissed and break the contact in the encounter" (p. 286)
		Afrell[43]: Although certain expressions of emotions were difficult for physiotherapists to respond to, ultimately physiotherapists found that these emotional displays allowed them **to better understand, and thus, treat their patients.**	"Expression of feelings in the clinic facilitates the physiotherapist's understanding of the patient's problems and **influences the result of the treatment positively**" (p. 286)
(4) Collaborative communication	Seeking common ground	Øien[52]: Physical therapists cited the importance of checking in (both verbally and nonverbally) with patients to ensure they were on the same page, particularly with regard to the patient's experience of his/her body during movement. Understanding and 'common ground' were cited as key to **quality therapeutic care**.	"The physiotherapists described the importance of seeking for common ground of understanding. They aimed at exploring connections between tasks and goals of the treatment and being sensitive to the patient's need of readjustments. They said that they continually checked if verbal and nonverbal messages were interpreted in accordance with the intention of either the physiotherapist or the patient. The physiotherapists were constantly sensitive to the patient's experiences of her/his body." (p. 55)
			"In the clinical encounters, over time and to different extent, the physiotherapists and the patients contributed and gradually found common ground. This process was based on the exploration of movements and appeared as tentative or manifest negotiations between the participants. Their comments on the task and/ or the relationship seemed to **facilitate the collaboration**." (p. 55)
	Recognizing the importance of bidirectional communication	Serpanou[50]: Physical therapists recognized that bidirectional communication was necessary for **understanding and delivering the best treatment**.	"good communication with the patient helps in overcoming obstacles. Communication is a two-way process between the physical therapist and the patient, and it is necessary in order to **understand each other's problems and expectations**" (p. 38)
		Cooper[44]: Patients viewed the communication as "good" if they were involved, but they did not have to make treatment decisions (SDM) to feel this way.	"Good communication was related to participants feeling involved in the physiotherapy process, and participants who were not able to discuss their needs or who received poor explanations did not feel involved. According to the participants, good communication involved: taking time over explanations; using appropriate terminology; listening, understanding and getting to know the patient; and encouraging the patient's participation in the communication process"" (p. 247)
			"participants were happy for the physiotherapist to make most decisions as long as they were accompanied by good explanations" (p. 248)

Preferred Communication Themes	Communication Behavior Subthemes	Data Contributing to Subthemes	Direct Quotations Describing Communication Behaviors
	Collaboratively agreeing on treatment goals	Cowell[45]: Patients used the term "collaboratively agreeing treatment goals" which when applied, means the same as "shared decision making".	"Physical therapists used the term 'collaboratively agreeing treatment goals' which basically means 'shared decision making' " (p. 116)
(5) Professional accountability communication	Admitting limitations in professional skills or knowledge	Gard[41]: Admitting your limitation was considered an essential part of being a **good physical therapist**.	"Not being afraid to admit your limitations and ask for help if it is needed. … Not feeling ashamed about not knowing everything (5:5)[a]." (p. 235)
		Pincus[49]: Admitting limitations for the greater good of the patient was considered an important practice.	''. . .there is somebody who has 10 years more experience than me, why you don't go and see them. There are so many therapies about'' (p. 72)
	Asking for help when needed	Gard[41]: Asking for help was considered an important practice for physical therapists so that they could **deliver the best treatment possible**.	"It is important not to be afraid to admit your limitations [knowledge] and ask for help if it is needed (5:5)[a]" (p. 235)
(6) Informative communication	Reconciling verbal communication with information from nonverbal communication cues during movement	Askew[33]: Physical therapists would compare nonverbal pain behaviors during posture and movement with verbal accounts of the pain experience to **determine a course of therapeutic action**."	"if I feel that someone is not being honest with me in regards to the pain, often times I will discontinue to use modalities and pursue more of a therapeutic exercise and work hardening program in order to help me determine the patient's true level of pain and function ability" (p. 47)
			"The theme of non-legitimate pain as perceived by physical therapists is reflected in the following: '…where their pain pattern does not appear to be appropriate for the type of injury or diagnosis they have, it influences how we treat them. We may ask them to work through their perceived pain differently than somebody else or we may ask them to go back to work despite their subjective complaints of pain' " (p. 47)
			"the biggest tickler would be a patient's presentation to the clinic, if they are displaying any type of muscle guarding, antalgic gait pattern, taking a look at how they position them-selves, their overall posture, and then correlating your observational skills to what the patient is telling you as far as descriptions of their pain, frequency of pain, their tolerance of pain, looking at the onset of their pain and then also getting a pain rating and see how that all correlates" (p. 44)
	Providing education and information	Del Baño-Aledo[46]: Information provided frequently and openly with patients was preferred.	"When physiotherapists were willing to provide patients with regular information and education about their plan of care or health status, this had a positive impact on **patients' perceptions of service quality**." (p. 77)
		Cooper[44]: Patients found that information sharing amongst themselves (patients with the same condition) was helpful and preferred in addition to receiving information from their physical therapists.	"Some limitations of group rehabilitation have been presented; however, the most recurrent positive comment was that information sharing (particularly with other CLBP patients) was the most helpful part and should be done more frequently" (p. 248)

Preferred Communication Themes	Communication Behavior Subthemes	Data Contributing to Subthemes	Direct Quotations Describing Communication Behaviors
		Cowell[45]: Having a trusting relationship allowed physical therapists to debunk incorrect beliefs and **improve treatment** for their patients.	"a trusting relationship facilitating patient disclosure...providing opportunities to reconcile patient unhelpful beliefs and **enhancing adherence**" (p. 116)
	Reconciling patient perspectives	Synnott[51]: Physical therapists gave advice on the importance of having confidence in challenging their patients' false beliefs and to face conflict with direct communication.	"More specifically, many participants described how they were now confident to challenge the patient's belief system, even if this led to some conflict. Now I'm more inclined to say 'Listen, hold on a minute. Anyway I've just got to re-examine your point of view on this' and that can sometimes lead to conflict... but I think you sometimes need conflict for conceptual change. (P1)" (p. 219)
		Cowell[45]: Investigators used the phrases "educating and developing patient insights" and "effective communication" synonymously.	"educating and developing patient insights…effective communication" (p. 116)
(7) Agenda-setting communication	Using summaries, 'yes/no' questions, and 'okay' to shift to a new topic of conversation	Opsommer[48]: Physical therapists were able to close down a pain conversation initiated by patients by summarizing the consultation and including a prognosis.	"summarize the consultation and to include a prognosis of the patient's problem" (p. 218)
		Opsommer[48]: Physical therapists were able to close an encounter using 'yes/no' formulations.	"…whereas he [physiotherapist B] finished the session with a final question 'did I forget to ask you a question?' The 'final concern' described in medical literature was observed with questions eliciting either 'yes' or 'no'-type responses and a preference for 'no'-type responses allowing a shift into closing of the encounter" (p. 218)
		Opsommer[48]: 'Okay' with a rising pitch was used by a physical therapist to invite agreement by their patient.	"The patient in Extract 1.1 is invited to agree to the proposed plan ('okay' with a rising pitch)." (p. 208)
(8) Meta-communication	Engaging in communication about communication	Afrell[43]: Patients were encouraged to talk about talking about pain which led to further understanding of patients' pain by both patients and physical therapists.	" 'If pain could talk, what would it say?' – an aspect of metacommunication was introduced, where patient and physiotherapist together reflected, giving new perspectives on the experience of pain." (p. 286)
		Øien[52]: Both patients and physical therapists alike would comment on "demanding moments of collaboration" and contributed to **improved therapeutic alliance**.	"Part process analysis includes communication at three levels: Theme, meta-communication and definition of the relationship. The theme is often verbally communicated. The meta-communication encompasses nonverbal emotional expressions related to the theme. The theme and the meta-communication structure and define the relationship (33–35)." (p. 55)
		Øien[52]: Both patients and physical therapists alike would comment on "demanding moments of collaboration" and contributed to **improved therapeutic alliance**.	"Communication about communication, appeared as means to open locked dialogues, and was initiated by the physiotherapists and the patients." (p. 55)

Secondary outcomes (highlighted in bold) include any intermediary or terminal outcome cited as being impacted by communication and potentially linked to patient outcomes.
[a]Refers to participant number 5 verbally expressing fear 5 times over the course of data collection.
Abbreviations: CFT= Cognitive functional therapy, SDM= Shared decision making, CLBP= Chronic low-back pain; P1=Participant 1